

74 - Client side validation

В части 42-43 разглеждахме server-side validation, при които имаме атрибути като [Required], [StringLength]... забавящи обработката и връщането на данните към клиента с милисекунди. Това може да бъде спестено с проверка в локалната машина на потребителя.

ASP.NET Core Server Side Validation

The screenshot shows a 'User Login' form with two input fields: 'Email' and 'Password'. Both fields have red error messages below them: 'The Email field is required.' and 'The Password field is required.' respectively. A 'Remember me' checkbox and a 'Login' button are at the bottom. To the right, a code block shows the C# class `LoginViewModel` with `[Required]` attributes on the `Email` and `Password` properties.

```
public class LoginViewModel
{
    [Required]
    public string Email { get; set; }

    [Required]
    public string Password { get; set; }
}
```

`if (ModelState.IsValid)` - Breakpoint

ASP.NET Core Client Side Validation

jquery.js

jquery.validate.js

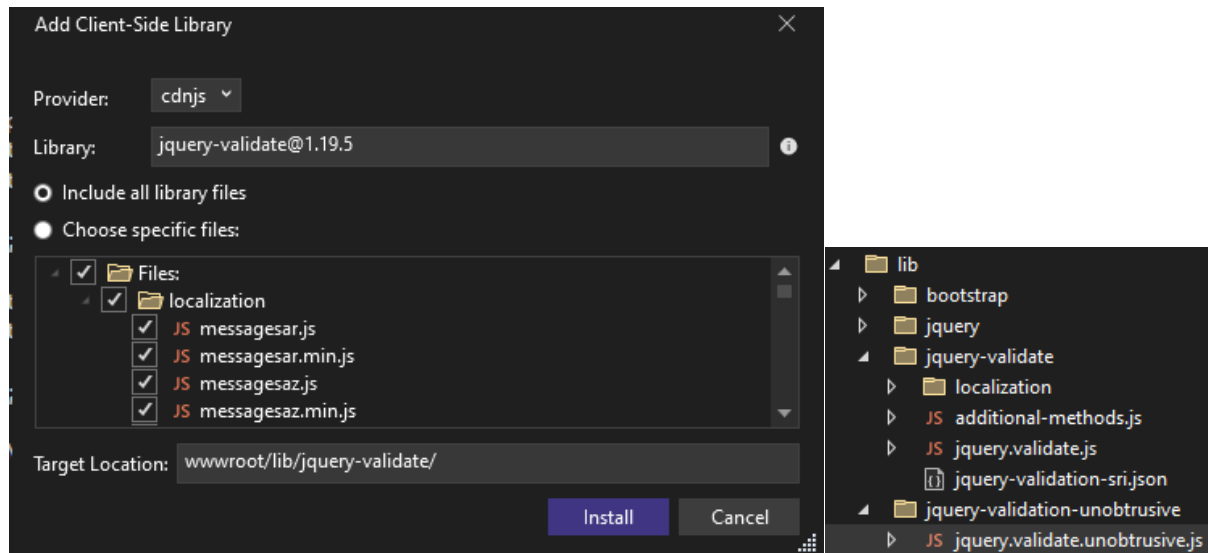
jquery.validate.unobtrusive.js

- You may use **Library Manager** to install client side libraries
- Library Manager : **Part 34 of ASP.NET Core tutorial**

Към момента имаме jquery, остава да инсталираме и останалите.

```
oller.cs | _Layout.cshtml | AccountController.cs*
<link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>
```

Project folder -> Add -> Client-side library -> jquery-validate



D&D

```
oller.cs | _Layout.cshtml | AccountController.cs* | jquery.validate.unobtrusive.js
<link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/jquery-validate/jquery.validate.js"></script>
<script src="~/lib/jquery-validation-unobtrusive/jquery.validate.unobtrusive.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>
```

```

ntroller.cs  _Layout.cshtml  AccountController.cs*  jquery.validate.unobtrusive.js
EmployeeManagement.Controllers.AccountControl  Login(LoginViewModel model, string returnUrl)

[HttpPost]
[AllowAnonymous]
0 references
public async Task<IActionResult> Login(LoginViewModel model, string returnUrl)
{
    if (ModelState.IsValid)  BreakPoint
    {
        var result = await signInManager.PasswordSignInAsync(model.Email, model.Password,
                                                                model.RememberMe, false);
        if (result.Succeeded)
        {
            if(!string.IsNullOrEmpty(returnUrl)) // && URL.LocalURL(returnURL));
            {
                return LocalRedirect(returnUrl);
            }
            else
            {
                return RedirectToAction("index", "home");
            }
        }

        ModelState.AddModelError(string.Empty, "Invalid Login attempt");
    }
    return View(model);  on the local machine check
}

```

User Login

- The Email field is required.
- The Password field is required.

Email

The Email field is required.

Password

The Password field is required.

☐ Remember me

Login

User Login

- The Email field is required.
- The Password field is required.

Email

1

The Email field is not a valid e-mail address.

Password

The Password field is required.

☐ Remember me

Login

В този урок, още след въвеждане на първия знак в полето за парола, проверката тече автоматично на локалната машина и съобщението за грешка изчезва, съответно ако този знак се изтрие и полето отново стане празно, съобщението се появява. За подобни промени в реално време е важна подредбата:

```

<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/jquery-validate/jquery.validate.js"></script>
<script src="~/lib/jquery-validation-
unobtrusive/jquery.validate.unobtrusive.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>

```

```

public class LoginViewModel
{
    [Required]
    [EmailAddress]
    4 references
    public string Email { get; set; }

    [Required]
    [DataType(DataType.Password)]
    4 references
    public string Password { get; set; }

    [Display(Name = "Remember me")]
    4 references
    public bool RememberMe { get; set; }
}

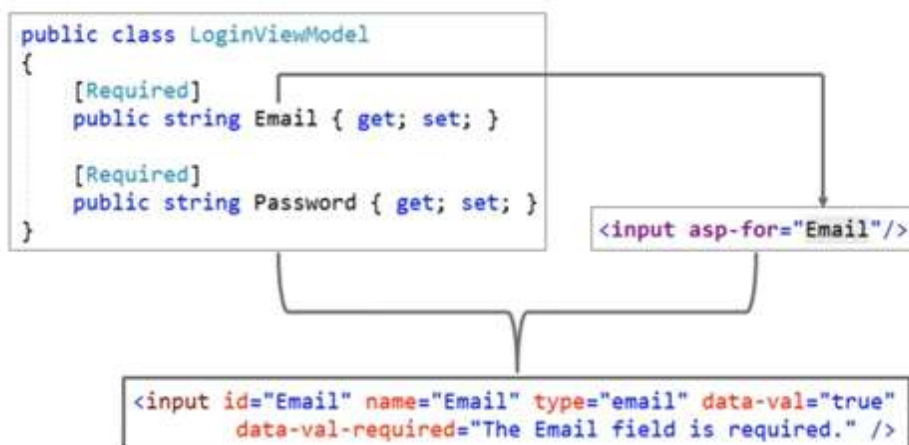
```

```

<div class="form-group">
  <label for="Email">Email</label>
  <input class="form-control" type="email" data-val="true" data-val-email="The Email field is not a valid e-mail address." data-val-required="The Email
  <span class="text-danger field-validation-valid" data-valmsg-for="Email" data-valmsg-replace="true"></span>
</div>

```

data-val attributes



➤ **jquery.validate.unobtrusive.js** reads **data-val** attributes and performs the client side validation.


```
Startup.cs | HomeController.cs | Layout.cshtml* | AccountController.cs | jquery.validate.unobtrusive.js
<environment include="Development">
  <link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
  <script src="~/lib/jquery/jquery.js"></script>
  <script src="~/lib/jquery-validate/jquery.validate.js"></script>
  <script src="~/lib/jquery-validation-unobtrusive/jquery.validate.unobtrusive.js"></script>
  <script src="~/lib/bootstrap/js/bootstrap.js"></script>
</environment>
<environment exclude="Development"> Staging, Production...
  <!-- CSS only -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvAphHNL/vI1Bx"
    crossorigin="anonymous"
    asp-fallback-href="~/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only"
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">
  <script src="~/lib/jquery/jquery.slim.min.js"></script> No validation
  <script src="~/lib/bootstrap/js/bootstrap.esm.min.js"></script>
</environment>
```

Client Side Validation - Not Working

- Make sure browser support for JavaScript is **NOT disabled**
- Make sure the client-side validation libraries are **loaded in the order specified**

```
<script src="jquery.js"></script>
<script src="jquery.validate.js"></script>
<script src="jquery.validate.unobtrusive.js"></script>
```

- Make sure the validation libraries are loaded **for the environment you are testing against**. Development, Staging, Production etc.